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REVIEW PAPER

Change Agent Training in Behavior Analytic Procedures for People with Developmental and Intellectual Disabilities: A Meta-Analysis

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Abstract

This paper reports a meta-analysis to evaluate procedures to train change agents in discrete trial teaching, preference assessment, and mand training. Behavioral skills training packages were the most commonly used and were consistently and highly effective in changing change agent behavior. Procedural variation in training overlapped in effectiveness. Training was slightly less effective for parents than for staff and university students. The effect size for the behavior of the individuals with whom the change agents applied behavioral procedures was slightly smaller than for change agents and was reported in only about one third of papers. Behavioral skills training is a highly effective and robust approach to delivering change agent training for different skills training procedures.

Keywords Change agent · Behavioral skills training · Training · Developmental disabilities

The costs of treatment for individuals with developmental disabilities (DD) can be quite high. For example, lifetime costs for people with intellectual disabilities (ID) who were born in 2000 were estimated at $51.2 billion US dollars in 2003 (CDC 2004). These estimates did not include treatment for individuals with autism spectrum disorders (ASD) or additional expenses like hospital outpatient visits, emergency room visits, residential care, and family out-of-pocket expenses. The actual economic costs of treating individuals with intellectual disabilities may, therefore, be even higher. Ganz (2006) estimated societal costs for individuals diagnosed each year with autism at 3.2 million US dollars over their lifetimes. Effective treatment for individuals with ASD can reduce lifetime costs by up to 65% (Järbrink and Knapp 2001). Much of these costs related to the costs of paid staff and lost family income. Thus, training change agents to use effective procedures to deliver treatments that work is important. This training not only helps change agents to avoid exposing learners to harmful or ineffective treatments, but also ensures that benefits are maximized and costs are minimized.

Authors of several recent meta-analyses have evaluated the single-subject literature to determine the most effective interventions for people with developmental disabilities (Ganz et al. 2012; Kokina and Kern 2010; Mason et al. 2013; Reichow and Wolery 2009; Vanderkerken et al. 2013; Wong et al. 2015) and have identified applied behavior analysis as an evidence-based practice for individuals with autism; however, even highly efficacious treatments are unlikely to be implemented if few people in clinical practice are competent to provide them (Chambless and Hollon 1998). Chambless and Hollon (1998) and Horner et al. (2005) agreed that procedural integrity should be closely monitored through training and monitoring based on actual samples of change agent behavior.

Researchers have used meta-analyses to assess the aspects of effectiveness in training change agents, including therapeutic staff and family members, to apply interventions with people with DD (Brock and Carter 2016; Brock et al. 2017; Kaminski et al. 2008; Van Oorsouw et al. 2009). They have evaluated specific treatment practices for positive parenting (Dyches et al. 2012), the Picture Exchange Communication System (PECS) (Tien 2008), and applications to ameliorate certain types of problems, such as externalizing behaviors and disruptive behavior disorders (Maughan et al. 2005) and for training certain types of change agents such as parents (Roberts 2011), peers (Zhang and Wheeler 2011), and

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university students (Fazzio et al. 2009; Thiessen et al. 2009). Although these meta-analyses have added to the existing research into change agent training, they have several shortcomings. Specifically, authors that have completed meta-analyses on change agent training have not: (a) evaluated study quality using criteria such as those proposed by Horner et al. (2005); (b) calculated effect sizes using multiple procedures to determine, both visually and mathematically, which are the most effective change agent training techniques for both change agents and people with DDs; and (c) aggregated data on training different types of change agents across multiple behavior analytic procedures to determine whether similar types of training lead to similar outcomes across change agents and procedures. Therefore, the purpose of the current meta-analysis was to address these shortcomings. Specifically, research studies in change agent training on three common procedures in applied behavior analysis: (a) discrete trial teaching (DTT); (b) preference assessment (PA); and (c) mand training (MT) were evaluated. This evaluation was completed to determine which change agent training packages were most effective in disseminating the above skills by calculating individual, mean, and omnibus effect sizes and by aggregating and disaggregating data across procedure types, training types, and change agent types.

Method

Search Procedures

Literature Searches Database searches were completed for PSYCinfo©, PubMed©, and ERIC© using the terms: discrete trial*, preference*, assessment*, and mand. These procedures were selected because (1) change agents commonly use them in both home- and center-based programs for people with DD and (2) they are well established as evidence-based practices for people with ASD. The National Professional Development Center (NPDC) on ASD recognized DTT as one of 27 evidence-based practices (EBP) for people with ASD. Similarly, NPDC identified preference assessment as part of antecedent-based intervention and mand training as a part of functional communication training (FCT) (National Professional Development Center 2014). Searches were limited to include only articles published in peer reviewed journals. The PubMed search was limited to include only human research results. There were 1078 papers, including 871 from PSYCinfo©, 100 from PUBmed©, and 107 returns from ERIC© for DTT; 1133 results, including 847 from PSYCinfo©, 41 from PUBmed©, and 245 returns from ERIC© for PA; and 664 results, including 345 from PSYCinfo©, 160 from PUBmed©, and 159 returns from ERIC© for MT.

Inclusion Criteria Two screeners read the headings and abstracts of retained papers and coded the articles as either "1," indicating that the article described a study addressing change agent training in one of the three behavior analytic procedures, or "2," indicating they did not.

A change agent was an individual responsible for providing care to ensure the health, safety, or educational instruction of a person with a developmental disability. Most often change agents in included articles were educational staff from schools or hospitals or family members. In some studies, university students acted as change agents if their course of study included providing educational instruction to people with developmental disabilities. Agreement between screeners was calculated on 100% of the returns by dividing the total number of agreements by the total number of agreements plus disagreements and multiplying by 100. After elimination of duplicate papers, screeners applied the following inclusion criteria: (a) the study was an experiment using a reversal, alternating treatments, multiple baseline design (MBD), multiple probe design (MPD), or a changing criterion design; (b) participants instructed individuals with a DD or traumatic brain injury (TBI), confederates role playing people with DD or TBI, or used computer program simulations of people with DD or TBI; and (c) the study included observational data on instructor accuracy of implementation of the behavior analytic procedure as a dependent variable.

Descriptions of Retained Articles Coders extracted data from each retained article on (a) authors; (b) year of publication; (c) type of training; (d) number of change agent participants trained; (e) type of change agent participants trained (staff, family members, or university students); (f) number of learner participants with ID, DD, or a TBI; (g) learner participants' diagnoses, ages, and sex; and (h) whether the authors reported learner participant data.

Categorizations of training types were necessary in order to compare the independent variables to which change agents were exposed. For example, if an article's authors reported using instructions, modeling, rehearsal, and feedback to train change agents, the coders categorized the independent variable as *BST*. If authors employed all four components of BST to train change agents remotely over a computer and never provided in vivo training, the coders recorded this as *computer-based BST*. If authors only required participants to watch video models of instructors performing the behavior analytic task with 100% accuracy coders recorded the type of training as *video-modeling*. If the authors required participants to watch and score videos of people performing the task with varying degrees of accuracy, the coders recorded this as *video monitoring*. If an article's authors used BST to train supervisors to implement the behavior analytic task and then the trained supervisors used BST to train their staff, the coders recorded this as *pyramidal BST*. Finally, the coders recorded

learner types exactly as the authors described, except that they coded "autism," "autism spectrum disorder," "autistic disorder," and similar diagnoses as "autism spectrum disorder."

Quality Indicator Scoring The Horner et al. (2005) quality indicators were modified to ensure reliability while maintaining their integrity. The first author and four doctoral ABA students scored 10 articles from a previous meta-analysis using the original Horner et al. (2005) criteria. All doctoral students read Horner et al. (2005) and completed 10 study questions before completing the first round of scoring. When interobserver agreement (IOA) was less than 80%, all doctoral students met and discussed quality indicators that may have required modification to increase IOA, agreed upon modifications, and scored an additional 10 articles upon leaving the meeting. There were three scorer meetings, resulting in several modifications to the Horner et al. (2005) criteria. Upon the fourth round of scoring, IOA for all of the articles was at least 80%. The version of the quality indicators used for this meta-analysis is summarized in Table 1. The quality indicators contained 20 items in seven domains.

The raters scored each retained article using the modified Horner et al. (2005) indicators. Scorers inserted a plus or minus into a cell next to each indicator on an Excel© spreadsheet. An independent doctoral student also independently scored 30% of the articles across all behavior analytic procedures according to the final modified version of the Horner et al. (2005) indicators. The overall IOA was 98.5% (range 90–100), and IOAs were 95.6% (range 90–100%), 100%, and 100% for the DTT, PA, and MT articles, respectively.

Data Extraction The individual participants served as the units of analysis. Percent of non-overlapping data (PND) (Scruggs et al. 1987) and improvement rate difference (IRD) (Parker et al. 2009) calculations were completed for 195 participants (169 change agents and 26 learners) from 32 studies. The data were then aggregated and disaggregated to compare effect sizes across and between the type of change agents trained, the type of training completed, and the behavior analytic procedure trained. IRD confidence intervals were also calculated to determine the reliability of and differences between effect sizes and omnibus IRD values.

Free online software (http://www.vassarstats.net/prop2_ind.html) was used to calculate IRD confidence intervals, and WinPepi (http://www.biomedcentral.com/1742-5573/content/1/1/6) was used to calculate omnibus IRDs (Parker et al. 2011). Omnibus IRDs reflect an aggregation of data points across participants or experiments rather than a mean and, in effect, take the data from separate individuals and calculate an overall IRD as if all baseline and intervention proportions for each individual were part of one larger experiment. The output confidence interval was narrower than for

Table 1 Quality indicators for single-subject research used for scoring

Description of participants and setting

- Participants are described with sufficient detail to allow others to select individuals with similar characteristics: age, sex, and disability diagnosis are given.
- The process for selecting participants is described with replicable precision: any inclusion or exclusion criteria are listed (e.g., only certified special education teachers were selected as participants, teachers who were board certified behavior analysts were not selected).
- Critical features of the physical setting are described with sufficient precision to allow replication: basic type of setting is given (e.g., school, home, hospital)

Dependent variable

- Dependent variables are described with operational precision.
- Each dependent variable is measured with a procedure that generates a quantifiable index.
- Measurement of the dependent variable is valid and described with replicable precision. Validity = the experimenters measure the parameter of the dependent measure that they purport to measure (e.g., sound measured as decibels, occurrence/nonoccurrence of behavior measured as frequency, rate, duration)
- Dependent variables are measured repeatedly over time.
- Data are collected on the reliability or interobserver agreement associated with each dependent variable.
- IOA levels meet minimal standards {e.g., IOA average = 80%; Kappa = 60%).

Independent variable

- Independent variable is described with replicable precision.
- Independent variable is systematically manipulated and under the control of the experimenter.
- Fidelity of implementation (procedural integrity/treatment fidelity) is measured for the independent variable.

Baseline

- Baseline phase provides repeated measurement of a dependent variable and establishes a pattern of responding that can be used to predict the pattern of future performance, if introduction or manipulation of the independent variable did not occur: typically requires multiple data points (five or more, although fewer data points are acceptable in specific cases) without substantive trend, or with a trend in the direction opposite that predicted by the intervention, a trend in the direction predicted by the intervention is not acceptable. Less than 3 data points in baseline is not acceptable.
- Baseline conditions are described with replicable precision.

Experimental control/internal validity

- The design provides at least three demonstrations of experimental effect at three different points in time. See Horner et al., (2005), p.168–170
- The design controls for common threats to internal validity (e.g., permits elimination of rival hypotheses).
- The results document a pattern that demonstrates experimental control.

External validity

- Experimental effects are replicated across at least 3 participants, target responses, settings, or materials to establish external validity. Which of these will be counted will depend upon the experimental design (e.g., MBD across skills, vs MBD across participants)

Social validity

- Social validity is enhanced by implementation of the independent variable by typical intervention agents, in typical physical and social contexts (e.g., teachers in schools, parents in home)
- Social validity data were collected (questionnaire, rating scale, etc.)

any of the individually entered IRDs because of the increased quantity of data from the combined individual IRDs. Effect sizes were then compared across the following variables: (a) participants' overall training effects; (b) participants' overall generalization effects; (c) participants' overall maintenance effects; (d) learner training effects; (e) learner generalization effects; (f) learner maintenance effects; (g) change agent overall training effects; (h) change agent training effects for each procedure; (i) change agent training effects for each independent variable; (j) training effects for participants of a certain change agent type; (k) change agent training effects for studies that met 80% or more of the quality indicators; (l) change agent training effects for studies that met less than 80% of the quality indicators; (m) change agent training effects for studies where experimenters took procedural integrity data; (n) change agent training effects for studies where experimenters did not take procedural integrity data; (o) change agent training effects for studies where experimenters collected social validity data; and (p) change agent training effects for studies where experimenters did not collect social validity data.

Interobserver Agreement A random number generator was used to select 30% of the retained studies for each meta-analysis to evaluate IOA. A doctoral student calculated individual PNDs and IRDs. IOA was calculated by dividing the total number of agreements by the sum of agreements plus disagreements and multiplying the number by 100%. Equal scores were instances where both scorers entered the same notation for the same participant. Disagreements were instances where observers entered different scores for the same participant. IOAs were 100, 100, and 91.7% for the DTT, PA, and MT meta-analyses, respectively.

Results

General Article Descriptions

Table 2 presents detailed descriptions of retained articles. The most represented journal was *The Journal of Applied Behavior Analysis* followed by *Research in Autism Spectrum Disorders*, *Research in Developmental Disabilities*, *Education and Training in Autism and Developmental Disabilities*, and *Focus on Autism and Other Developmental Disabilities*. The most represented publication year was 2012, followed by 2008, 2009, 2007, 2011, and 2013. The most common experimental design was standard MBD followed by modified MBD, modified MPD and standard MPD. There were 18 total types of change agent training employed. The most represented training type was BST, followed by computer-based BST, instructional manuals, video modeling, video monitoring, and an instructional manual in combination with a video model. All other independent variables were used only once. The most represented types of change agents were staff members, followed by parents and university students. The most represented learner diagnosis was ASD followed by PDD, intellectual disabilities (mental retardation), Down syndrome, cerebral palsy, nonspecific behavior disorder, multiple genetic anomaly, and expressive language delay. The most common learner age group was young children, followed by adolescents, middle aged adults, and teens. Article authors did not describe the specific ages of the learners with whom the change agent participants typically interacted in nine articles. Most learners were male. Authors did not describe the sex of the learners with whom the change agent participants interacted in 19 articles. Authors of only nine articles reported learner data in which the most represented group was young male children with ASD, followed by young male children with intellectual disabilities. Although authors of nine articles reported learner data for a total of 33 learners, authors of only eight of these articles presented learner data on figures to allow for IRD and PND calculation. One article's authors presented data for six learners on a figure, but only implemented the independent variable with five. The same article's authors presented data for six change agents on a figure, but only implemented the independent variable with five. Therefore, IRDs and PNDs were calculated for a 169 change agents across 32 studies and 26 learners across eight studies.

Quality Indicator Scores

Table 3 shows the quality indicator data for all retained articles. All retained articles met at least 13 quality indicators. There was little uniformity as to which indicators were met, except that all studies met the quality indicators for the general area of participants, and settings.

Effect Sizes

Effects by Procedure Type The mean PND for DTT training effects was 96.1% (range 27–100%; *highly effective*) for 96 change agents across 23 studies. The mean IRD was 0.98 (range 0.65–1), and the omnibus IRD was 0.95 CI_{95} (0.93, 0.96; *large effects*). The mean PND for generalization of DTT training effects was 94.8% (range 0–100%; *highly effective*) for 55 change agents across 13 studies. The mean IRD was 0.96 (range 0.36–1) and the omnibus IRD was 0.96 CI_{95} (0.92, 0.99; *large effects*). The mean PND for maintenance of DTT training effects was 99.6% (range 88–100%; *highly effective*) for 34 change agents across eight studies. The mean IRD was 0.99 (range 0.88–1), and the omnibus IRD was 0.99 CI_{95} (0.94, 1; *large effects*).

The mean PND for PA training was 96.7% (range 50–100%; *highly effective*) for 61 change agents across six studies. The mean IRD was 0.98 (range 0.70–1), and the omnibus IRD was 0.99 CI_{95} (0.97, 1; *large effects*). The mean PND for

Table 2 Descriptions of retained articles from database searches for terms "Discrete Trial*," "Preference*Assessment*," and "Mand"

Authors	Design	Type of training	Number of change agent participants	Type of change agent participants	Number of learners	Learners—diagnosis	Learners—age	Learners—sex	Learner data reported
Discrete trial teaching									
Nosik et al. (2013)	MBD	BST, computer-based BST	6 (3 BST, 3 CBT)	Staff	1	ASD	39	Not given	No
Garland et al. (2012)	MBD	Computer-based BST	4	Staff	0	ASD	Not given	Not given	No
Young et al. (2012)	MBD	Study 1—manual and video Study 2—manual, role-playing, feedback, video, BST	10 (study 1 = 5, study 2 = 3, study 3 = 2)	Parents	Study 1 = 3, Study2 = 5	ASD	Study 1 = 5–6 Study 2 = 3–5	Study1 = m Study 2 = 2 m, 3 f	No
May et al. (2011)	MBD—modified with embedded reversal	BST vs BST + stimulus prompt	3	Staff	0	ASD	Not given	Not given	No
Thomson et al. (2012)	MBD—modified	Self-instructional manual, video model	8	Staff	0	ASD	Not given	Not given	No
Severtson and Carr (2012)	MBD—non-concurrent	BST	6	Staff	0	ASD	Not given	Not given	No
Vladescu et al. (2012)	MBD	Video modeling with voiceover instruction	3	Staff	3	ASD, GDD, PDD	2–5	Not given	No
Nosik and Williams (2011)	MBD	Computer-based BST	4	Staff	0	ID	Not given	Not given	No
Catania et al. (2009)	MBD	Video modeling with voiceover (feedback for one participant)	3	Staff	0	ASD	Not given	Not given	No
Williams and Gallinat (2011)	MBD	Video monitoring	4	Staff	1	Severe ID	43	m	No
Lafasakis and Sturmey (2007)	MBD	BST	3	Parents	3	ASD, MR, DS	4	m	Yes
Fazzio et al. (2009)	MBD—modified	BST	5	University students	3	ASD	2–4	Not given	No
Thiessen et al. (2009)	MBD—modified	A revised self-instructional manual	4	University students	1	ASD	2–6	Not given	No
Bolton and Mayer (2008)	MBD—modified	BST	3	Staff	3	ASD, PDD-NOS	1–5	m	No
Ward-Horner and Sturmey (2008)	MBD	BST and general-case training	3	Parents	3	ASD	4–5	m	Yes
Belfiore et al. (2008)	MBD	Video self-monitoring	3	Staff	Not given	ASD	Not given	Not given	No
Dib and Sturmey (2007)	MBD	BST	3	Staff	3	ASD	9–12	m	Yes
Downs et al. (2008)	MBD	BST	6	University students	4	ASD, PDD, GDD, and CP	3–5	2 m, 2 fm	Yes (0 in figure)
Sarokoff and Sturmey (2008)	MBD	BST	3	Staff	5	ASD	Mean of 5	Not given	Yes (3 in figure)
Leblanc et al. (2005)	MBD		3	Staff	3	ASD	5–9	m	No

Table 2 (continued)

Authors	Design	Type of training	Number of change agent participants	Type of change agent participants	Number of learners	Learners—diagnosis	Learners—age	Learners—sex	Learner data reported
		Abbreviated performance feedback							
Crockett et al. (2007)	MBD—modified	BST	2	Parents	2	ASD, severe MR	4	m	Yes
Sarokoff and Sturmey (2004)	MBD	BST	3	Staff	1	ASD	3	Not given	No
Salem et al. (2009)	MBD	Manual and video	4	University students	1	ASD	4	m	No
Preference assessment									
Graff and Karsten (2012)	MBD	Enhanced instructions	11	Staff	4	ASD, DD	3–15	Not given	No
Pence et al. (2012)	MPD—modified	BST, pyramidal BST	1) 6, 2) 18	Staff	12	Behavior disorders, LD, DD	Not given	Not given	No
Bishop and Kenzer (2012)	MPD—concurrent	BST—context difference	11	Staff	Not given	ASD	Not given	Not given	No
Roscoe and Fisher (2008)	MPD—modified	BST	8	Staff	0	DD	Not given	Not given	No
Roscoe et al. (2006)	MPD—modified	(1) Instructions, (2) instructions and feedback, (3) instructions and money, (4) instructions, feedback, and money	4	Staff	2	ASD, profound MR	4–12	m	No
Lavie and Sturmey (2002)	MBD	BST	3	Staff	8	ASD	3–5	Not given	No
Mand training									
McCulloch and Noonan (2013)	MBD	Online training video, self-evaluation, and self-modeling	3	Staff	3	ASD and developmental delays	6–10	2 m, 1 f	Yes
Mobayed et al. (2000)	MPD	Written instructions, oral presentation, rehearsal, and feedback	4	Parents	4	Multiple genetic anomalies; developmental and expressive language delays	2	m	Yes
Nigro-Bruzzi and Sturmey (2010)	MBD	BST	6 (5 IV)	Staff	6 (5 IV)	ASD	2–6	Not given	Yes (5 IV)
		Total change agent effect sizes	169	Total learner effect sizes:	26				

Table 3 Quality indicator scoring for all articles retained for the discrete trial meta-analysis

Article	Participants/ setting	Dependent variable	Independent variable	Baseline	Internal validity	External validity	Social validity
Discrete trial teaching							
1—Nosik et al. (2013)	X	X		X	X	X	
2—Garland et al. (2012)	X	X	X	X	X	X	
3—Young et al. (2012)	X	X	X			X	X
4—May et al. (2011)	X	X		X	X	X	
5—Thomson et al. (2012)	X	X	X	X		X	
6—Severtson and Carr (2012)	X	X		X	X	X	X
7—Vladescu et al. (2012)	X			X	X	X	
8—Nosik and Williams (2011)	X	X		X		X	
9—Catania et al. (2009)	X	X		X	X	X	
10—Williams and Gallinat (2011)	X	X					
11—Lafasakis and Sturmey (2007)	X	X		X	X	X	
12—Fazzio et al. (2009)	X	X	X	X	X	X	X
13—Thiessen et al. (2009)	X	X	X	X	X	X	X
14—Bolton and Mayer (2008)	X	X	X			X	
15—Ward-Horner and Sturmey (2008)	X	X			X	X	
16—Belfiore et al. (2008)	X	X	X	X	X	X	
17—Dib and Sturmey (2007)	X	X		X	X	X	
18—Downs et al. (2008)	X	X		X	X	X	
19—Sarakoff and Sturmey (2008)	X	X		X	X	X	X
20—Leblanc et al. (2005)	X	X		X	X	X	X
21—Crockett et al.(2007)	X	X					
22—Sarakoff and Sturmey (2004)	X	X		X	X	X	
23—Salem et al. (2009)	X	X	X	X	X	X	X
Preference assessment							
1—Graff and Karsten (2012)	X	X	X	X	X	X	
2—Pence et al. (2012)	X	X	X		X	X	
3—Bishop and Kenzer (2012)	X	X			X	X	
4—Roscoe and Fisher (2008)	X	X			X	X	
5—Roscoe et al. (2006)	X	X		X	X	X	
6—Lavie and Sturmey (2002)	X	X		X	X	X	
Mand training							
1—McCulloch and Noonan (2013)	X	X		X	X	X	
2—Mobayed et al. (2000)	X	X		X	X	X	X
3—Nigro-Bruzzi and Sturmey (2010)	X	X		X	X	X	

generalization of training effects was 100% (highly effective) for 21 change agents across three studies. The mean IRD was 1, and the omnibus IRD was 1 CI_{95} (0.97, 1; *large effects*). The mean PND for maintenance of training effects was 100% (highly effective) for 11 change agents included in one study. The mean IRD was 1 and the omnibus IRD was 1 CI_{95} (0.85, 1; large effects).

The mean PND for MT training effects was 100% (*highly effective*) for 12 change agents across three studies. The mean IRD was 1, and the omnibus IRD was 1 CI_{95} (0.92, 1; *large effects*). The mean PND for generalization of training effects

was 100% (*highly effective*) for five change agents included in one study. The mean IRD was 1 and the omnibus IRD was 1 CI_{95} (0.77, 1; *large effects*). The mean PND for maintenance of training effects was 100% (*highly effective*) for seven change agents across two studies. The mean IRD was 1, and the omnibus IRD was 1 CI_{95} (0.86, 1; *large effects*).

Effects by Type of Change Agent Training Figure 1 presents the IRDs and 95% CIs in a forest plot for each of the 18 independent variables to which change agents were exposed. The aggregated mean PND was 95.9% (range 27–100%; *highly effective*) for 169 change agents across 32 studies. The mean IRD was 0.98 (range 0.65–1), and the omnibus IRD was 0.98 CI_{95} (0.97, 0.99; *large effects*). BST and pyramidal BST had the largest effect sizes and the narrowest confidence intervals. The mean PND for BST effects was 98.8% (range 27–100%; *highly effective*) for 62 change agents across 16 studies. The mean IRD was 0.98 (range 0.72–1), and the omnibus IRD was 0.97 CI_{95} (0.96, 0.99; *large effects*). The mean PND for pyramidal BST was 98.8% (range 83.3–100%; *highly effective*) for 24 change agents included in one study. The mean IRD was 0.98 (range 0.72–1), and the omnibus IRD was 0.97 CI_{95} (0.96, 1; *large effects*).

Effects by Type of Change Agent The aggregated mean PND for change agents was 97.4% (27–100%) for 169 change agents across 32 studies. The mean IRD was 0.95 (0.65–1), and the omnibus IRD was 0.97 CI_{95} (0.96, 0.98; *large effects*). The mean PND for staff member change agents was 98% (range 39–100%; *highly effective*) for 128 staff members across 23 studies. The mean IRD was 0.99 (0.65–1), and the omnibus IRD was 0.99 CI_{95} (0.98, 1; *large effects*). The mean PND for parent change agents was 94.2% (range 27–100%) for 22 parents across five studies. The mean IRD was 0.91 (0.67–1), and the omnibus IRD was 0.86 CI_{95} (0.83, 0.90; *large effects*). The mean PND for university student change agents was 100% (*highly effective*) for 19 university students across four studies. The mean IRD was 1, and the omnibus IRD was 1 CI_{95} (0.92, 1; *large effects*).

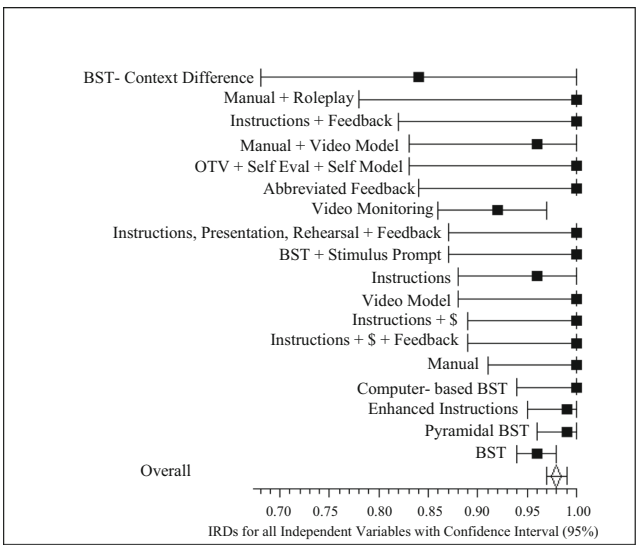


Fig. 1 The IRDs and confidence intervals (95%) for each of 18 independent variables (training types) across all three meta-analyses. The IRDs are ordered from the lowest to the highest lower limit of confidence from top to bottom. The white diamond crosses the X-axis at the omnibus IRD

Effects by Quality Indicators Met Figure 2 displays (a) IRDs for change agent participants from studies that met 80% or more of the quality indicators and from studies that met less than 80% of the quality indicators, (b) IRDs for change agent participants from studies where experimenters collected procedural integrity data and from studies where they did not, and (c) IRDs for change agent participants from studies where experimenters collected social validity data and from studies where they did not.

The mean PND for change agents included in studies that met 80% or more of the quality indicators was 98.2% (range 50–100%; *highly effective*) for 159 change agents across 29 studies. The mean IRD was 0.99 (range 0.67–1), and the omnibus IRD was 1 CI_{95} (0.97, 1; *large effects*). The mean PND for change agents included in studies that met less than 80% of the quality indicators was 71.8% (range 27–100%; *moderately effective*) for 10 change agents across three studies. The mean IRD was 0.88 (range 0.65–1), and the omnibus IRD was 0.83 CI_{95} (0.80, 0.87; *large effects*). The mean PND for change agents included in studies where experimenters collected procedural integrity data was 98.9% (range 67–100%; *highly effective*) for 80 change agents across 11 studies. The mean PND for change agents included in studies where experimenters did not collect procedural integrity data was 94.6% (27–100%; *highly effective*) for 89 change agents across 21 studies. The mean PND for change agents included in studies where the experimenters collected social validity data was 98.9% (range 67–100%; *highly effective*) for 55 change agents across 11 studies. The mean IRD was 0.99 (range 0.67–1), and the omnibus IRD was 1 CI_{95} (0.97, 1; *large effects*). The mean PND for change agents included in studies where experimenters did not collect social validity data was 95.6% (27–100%; *highly effective*) for 114 change agents across 21 studies. The mean IRD was 0.98 (0.65–1), and the omnibus IRD was 0.95 CI_{95} (0.93, 0.97; large effects).

Discussion

This meta-analysis addresses several shortcomings of previous meta-analyses and adds to the existing research on change agent training, by (a) assessing article quality, (b) using multiple measures of effect size, and (c) aggregating data on training different types of change agents across multiple behavior

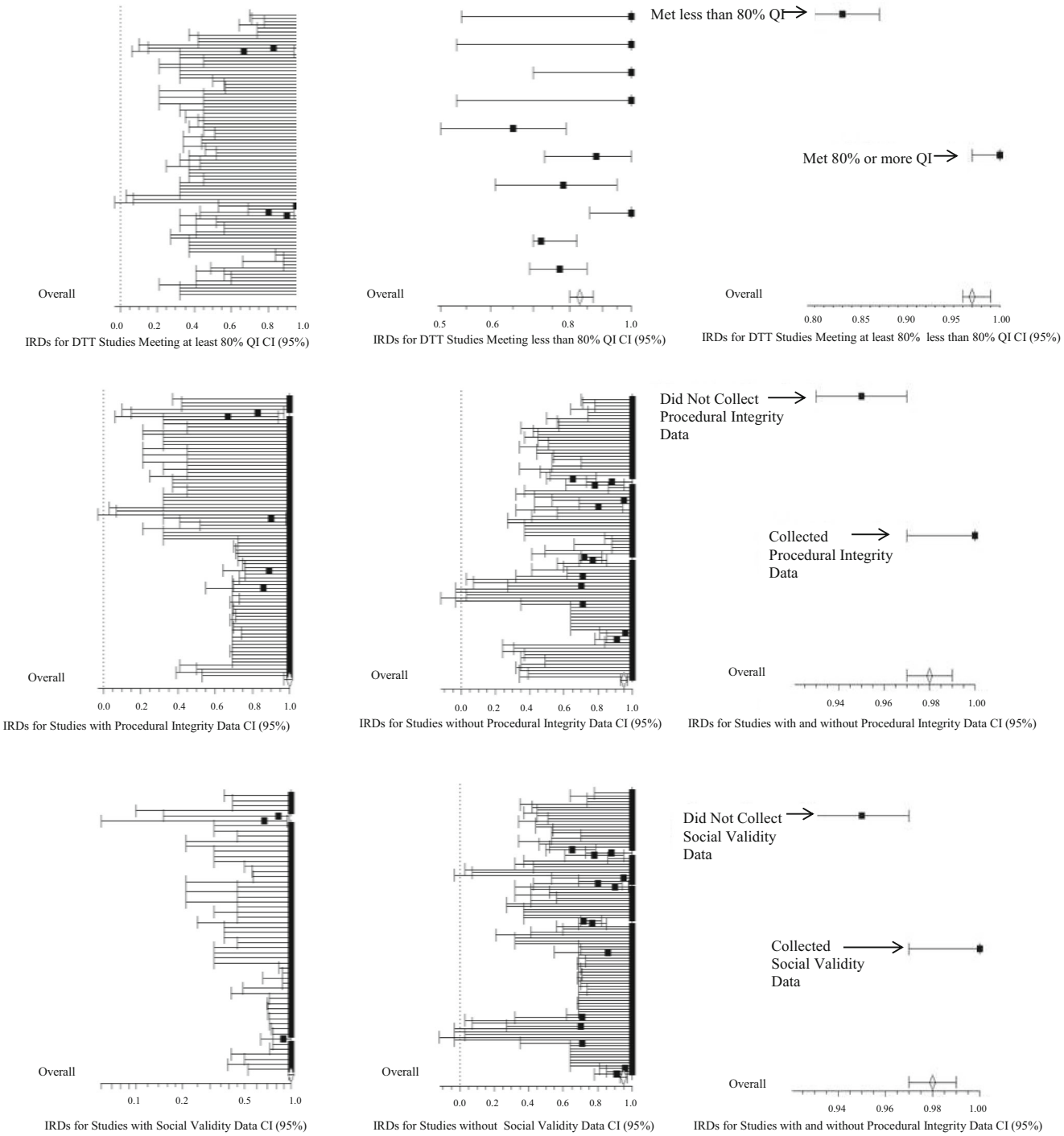


Fig. 2 IRDs and confidence intervals (95%) for DTT studies that met 80% or more of the quality indicators (top left panel) and that met less than 80% of the quality indicators (top center panel). The top right panel displays the omnibus IRDs for both. All studies included in the preference assessment and mand training meta-analyses met at least 80% of the quality indicators. IRDs and confidence intervals (95%) for studies where experimenters collected procedural integrity data (middle left panel) and for studies where they did not (middle center panel). The middle right panel displays the omnibus IRDs for both. IRDs and confidence intervals (95%) for studies where experimenters collected social validity data (bottom left panel) and for studies where they did not (bottom center panel). The bottom right panel displays the omnibus IRDs for both. The white diamond crosses the X-axis at the omnibus IRD for each forest plot

analytic procedures to determine whether similar types of training led to similar outcomes across change agents and caregiver training procedures. It also followed the traditional structure of meta-analytic research by the inclusion and compilation of descriptions of all included articles. By following the above processes, the results of this meta-analysis revealed

important information about the generalizability and quality of the current research in change agent training in behavior analytic procedures.

Generalizability of the Current Research Independent Variables

Training Types There were 18 types of change agent training. All training types were shown to be at least moderately effective, and most were highly effective in training change agents to implement all three ABA procedures; however, most of the training types only appeared in one to three studies. BST was by far the most well-represented training type with half of the included studies evaluating a BST package. Therefore, BST is currently the best evidenced for training change agents to implement these three ABA procedures. Several training types that may be valuable toward accomplishing efficient and cost-effective change agent training are currently under researched. For example, remote training packages including computer-based BST and online training videos with self-evaluation and self-models may potentially be quite beneficial in providing training to many change agents while requiring only minimal human resources once the initial training materials are developed and evaluated. Only three computer-based BST packages (which were slightly different in content) and one online training package were detected by the methods used in this meta-analysis.

Change Agents This review identified few types of change agents. Most studies evaluated staff member training, but very few evaluated training family members or university students. Further, nearly all of the parents trained in included studies were mothers; only one father was represented and no other members of extended families, such as grandparents, siblings, and nannies, were included. Overall, staff training was shown to be more effective than parent training, however; this distinction may have reflected the limited amount of parent training research evaluated rather than true difference.

Learner Data Authors of only 8 of 32 studies reported learner data, and when authors reported learner data, effects were smaller for learners than for change agents. This lack of learner data presents a potential problem with the social validity of the included studies. An argument for the validity of change agent training becomes more difficult to make without sufficient evidence that change agent training improves learner performance in meaningful ways. Smaller effect sizes for learners may reflect several relevant factors, such as lack of immediate child response during initial teaching programs, early extinction bursts, or problems with initial teaching programs (e.g., inappropriate consequences) that might be addressed quickly in the context of typical services. Future research should address this issue by including longer-term follow-up data on children and evaluation of more complex real-world caregiver training protocols that include procedures to address these issues.

Learner Type The range of learner types was limited. Although learners with ASDs were well represented and several other disability types appeared in the reviewed literature, most only appeared in one to four studies. The literature reviewed was also quite limited when it came to learner ages: Young children with disabilities were well represented, but there were only two of the eight studies where authors reported learner data that included an age group other than young children and these were adolescent participants; none were adults. Thus, one cannot conclude that change agent training is effective for those working with teenage or adult populations.

Research Quality The quality of the studies included in this meta-analysis was quite good: Almost all retained studies met at least 80% of the single-subject quality indicators, and better quality studies had larger effect sizes than studies with weaker designs. The most common limitations were that many studies did not collect procedural integrity or social validity data. Collecting procedural integrity data should be more common in studies where authors seek to improve the implementation of empirically supported therapies. In addition, the lack of social validity data, compounded by the scant amount of learner data reported, further weakens the argument for the social validity of change agent training. Although there were larger effect sizes for studies where authors collected procedural integrity and social integrity data versus those that did not, these differences were slight with nearly identical differences in PND and IRD confidence intervals that met at 0.97. Although it may be that the collection of these types of data may not necessarily lead to better results for participants, the lack of such data limits our understanding of the relationship between the independent and dependent variables. For example, the more variable learner outcomes might reflect differences in treatment integrity between children rather than other variables, but absent data on treatment integrity, it is not possible to know if this is the case.

Efficacy of Training Types There were few differences between training types: Nearly all were highly effective as all training types had large effects according to IRD calculations and all confidence intervals overlapped. Training types differed only by the sizes of their confidence intervals. BST had the smallest confidence interval due partly to the large number of participants. Therefore, although the training types performed similarly to each other across both effect size calculations, one can conclude with considerable confidence that BST was the best evidenced of the training types and that instructions alone, video monitoring, and BST where the training context differed from the follow-up data collection context were slightly less effective than the other caregiver training methods.

Limitations

One limitation of this meta-analysis was that it addressed only three behavior analytic procedures. Other procedures, such as more complex skills like using chaining and shaping, multi-component teaching packages with components spread out over time, and selecting goals accurately, may have smaller effect sizes than the three relatively simple teaching skills reviewed here. A second limitation was that calculation of PND and IRD effect sizes included training data. Many authors of systematic reviews and meta-analyses include only baseline and follow-up data in their calculations. Thus, the effect sizes reported here are more conservative than those reviews. Third, this meta-analysis included studies that employed modified versions of MBDs and MPDs, including concurrent MBDs and MPDs, which are not as strong as concurrent MBDs and MPDs; their inclusion in this meta-analysis could be questioned. Finally, this meta-analysis did not examine the contribution of components of training packages to identify the necessary, most effective, and efficient components. Future meta-analyses should include evaluation of the efficacy of training in a broader range of behavior analytic procedures, such as functional behavior assessment and analysis, data analysis, implementation of token economies, activity schedules, compliance training, shaping, chaining, and feeding interventions. In addition, methods of future meta-analyses might include a comparison of effect size calculations for studies that include training data with those that do not include such data.

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Compliance with Ethical Standards

Conflict of Interest The first author declares that she has no conflict of interest related to this article. The second author declares that he received speaking and consulting fees related to behavioral skill training which might represent a conflict of interest related to this article.

Human and Animal Rights and Informed Consent This article does not contain any studies with human participants or animals performed by any of the authors.

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